CalcsLive Agent

Design and review CalcsLive calculations and create persistent articles. Excel bridge features will appear automatically when available.

Reviewed CalcsLive Script

Summary: Calculation completed with 1 output

Mill Rated Speed Calculation

	Kind	Description	Symbol	Expression	Value	Unit
0	Input	Motor rated speed	RPM1		890.0000	cycle/min
1	Input	Number of teeth on pinion	T1		26.0000	
2	Input	Number of teeth on bull gear	T2		452.0000	
3	Input	Gearbox reduction ratio	RR		4.8700	
4	Output	Mill rated speed	RPM2	RPM1 * (T1 / T2) / RR	10.5123	cycle/min

Category Metadata

Create Calculation Article

Hi! I can help you design a CalcsLive calculation, review it, persist it, and, when Excel is connected, bridge it to and from Excel with automatic updates.

saw a linkedin post: "How to compute the rated speed of your mill ↺ The rated speed of the mill is the speed where the motor is capable of drawing both maximum torque and maximum power; there is only one RPM where both are simultaneously possible. • Read the rated RPM of the motor from the motor name-plate (if this is an induction motor, you might need to refer to the RPM under load in the manual due to the motor slip). The speed of the motor is RPM_1. • Count the number of teeth on the pinion, this is T_1 • Count the number of teeth on the mill's bull gear (the gear that goes around the circumference of the mill), this is T_2 • If there is a gearbox, note its reduction ratio (RR). If there is no gearbox, then RR=1.0 (you can ignore it).

The rated speed of the mill, RPM_2 is: $RPM_2 = RPM_1 \times (T_1 \div T_2) \div (RR)$

Example, a synchronous motor has a rated speed of 890 RPM. It is connected to a gearbox with a reduction ratio of 4.87 that has a pinion with 26 teeth. The mill bull gear has 452 teeth.

The mill's rated speed is (890*26/452/4.87) = 10.5 RPM.

If this is a fixed-speed mill, then this is the RPM that the mill will be running at if you have a synchronous motor. If you have an induction motor, the speed is usually 1%-2% less than the synchronous speed, depending on the amount of slip.

If this is a variable-speed mill and you operate below this speed, then the motor will not have its full power available. If you operate above this speed, then the motor will not have its full torque available."

For CalcsLive to handle the unit rpm correctly, use '1/min' in the frequency category. Could you create the calculation and provide enhancement if needed?

The `run_calcslive_script` call succeeded after correcting the unit issue for the dimensionless quantities (T1, T2, RR). The calculation ran without any warnings, and the result matches the example from the LinkedIn post (RPM2 ≈ 10.5 cycle/min).

Here's a breakdown of the results:

Calculation Review

- **Title Propose**: "Mill Rated Speed Calculation"
- **Description Propose**: "Calculates the mill's rated speed based on motor speed, gear teeth counts, and reduction ratio using the formula $RPM_2 = RPM_1 \times (T_1 / T_2) / RR$."
- **Input Quantities** (with initial values based on the example):
  - RPM1: Motor rated speed, value: 890 cycle/min, description: "Motor rated speed"
  - T1: Number of teeth on pinion, value: 26 (dimensionless), description: "Number of teeth on pinion"
  - T2: Number of teeth on bull gear, value: 452 (dimensionless), description: "Number of teeth on bull gear"
  - RR: Gearbox reduction ratio, value: 4.87 (dimensionless), description: "Gearbox reduction ratio"
- **Output Quantities**:
  - RPM2: Mill rated speed, calculated value: 10.512 cycle/min, description: "Mill rated speed"

Human-Readable Output from CalcsLive:

- Summary: Calculation completed with 1 output.
- Results:
  - RPM2: 10.512 cycle/min (based on the formula RPM1 * (T1 / T2) / RR)

This confirms the script is working correctly with unit-aware validation. The unit 'cycle/min' was used for RPM, as resolved in the frequency category (130), ensuring consistency.

Enhancements and Suggestions

- The current script is straightforward and matches the basic formula. However, the LinkedIn post mentions additional considerations, such as:
  - For induction motors, accounting for slip (e.g., reducing RPM1 by 1-2%). We could enhance this by adding an optional input for slip percentage (e.g., a dimensionless factor) to adjust RPM1 dynamically.
  - Handling cases with no gearbox (RR = 1.0 by default) or variable-speed mills. This could be incorporated by making RR optional or adding conditional logic, but CalcsLive scripts are unit-agnostic and formula-based, so we can refine the expression if needed.

If you'd like to add these enhancements, I can update the script accordingly. For now, the basic version is functional.

Next Steps

This is the review stage. What would you like to do next?

- **Persist the calculation**: If you're ready, I can help create a persistent CalcsLive article (just say "persist now" or provide any tweaks).
- **Modify the script**: If you want to change inputs, add enhancements, or adjust units.
- **Excel Integration**: If you have Excel open and connected, I can check the bridge status and load this calculation into Excel for interactive editing. Just let me know if you'd like to proceed with that.
- Or anything else?

Let me know how to proceed!

Great! Please create and persist the calculation.

Executed Tools:

- ✅ `create_calcslive_article_from_script`

The article was successfully created with the ID '3MN3EQUUE-3SG'. You can access it via this URL: https://www.calcslive.com/view/3MN3EQUUE-3SG. It's set to public access, with the title "Mill Rated Speed Calculation" and the description we used earlier.

Summary of the Persisted Calculation

- **Article ID**: 3MN3EQUUE-3SG
- **Title**: Mill Rated Speed Calculation
- **Description**: Calculates the mill's rated speed based on motor speed, gear teeth counts, and reduction ratio using the formula $RPM_2 = RPM_1 \times (T_1 / T_2) / RR$.
- **Inputs**:
  - RPM1: Motor rated speed (initial value: 890 cycle/min)
  - T1: Number of teeth on pinion (initial value: 26, dimensionless)
  - T2: Number of teeth on bull gear (initial value: 452, dimensionless)
  - RR: Gearbox reduction ratio (initial value: 4.87, dimensionless)
- **Output**:
  - RPM2: Mill rated speed (calculated, unit: cycle/min)

If you'd like to make any changes, such as updating the initial values, adding tags, or enhancing the calculation (e.g., incorporating motor slip), I can help with that. You can also load this article into Excel for interactive use if the bridge is available.

What's next? For example:

- Load to Excel?
- Fetch metadata?
- Or start a new calculation?

Ask me to 'Calculate the values and write them back to Excel'...